

bookcover.cls

Documentation — User Manual

L^AT_EX class for generating print-ready book covers

Compile with **LuaL^AT_EX** or **XeL^AT_EX**

v2.0 — 2026

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1 Installation

Place `bookcover.cls` in the same folder as your `.tex` file, or in any directory on your $\text{\LaTeX}$ search path. On Overleaf, simply upload the file to your project.

2 Class options

```
\documentclass[
  width      = 148,      % book page width in mm          (default: 148)
  height     = 210,      % book page height in mm         (default: 210)
  margin      = 10,      % content margin in mm; groove side gets
                        % margin + groove automatically (default: 10)
  bleed       = 20,      % bleed extension in mm          (default: 20)
  groove      = 8,       % binding hinge width in mm       (default: 8)
  paperwidth  = 420,     % output sheet width in mm        (default: 420)
  paperheight = 297,     % output sheet height in mm       (default: 297)
  pages       = 300,     % number of pages (sets spine width)
  grammage    = 75,      % paper weight in g/m2          (default: 75)
  colorFront  = 1B6B7B, % front panel colour, hex without #
  colorBack   = 1B6B7B, % back panel colour,   hex without #
  colorSpine  = D4A96A, % spine colour,         hex without #
]{bookcover}
```

2.1 Supported grammage values

Grammage (g/m ²)	Thickness per sheet
56	0.07 mm
60	0.08 mm
75	0.10 mm
90	0.12 mm
115	0.15 mm

3 Sheet anatomy

The full artwork is centred on the output sheet. Reading left to right:

| $\leftarrow$ bleed $\rightarrow$ | $\leftarrow$ back $\rightarrow$ | $\leftarrow$ groove $\rightarrow$ | $\leftarrow$ spine $\rightarrow$ | $\leftarrow$ groove $\rightarrow$ | $\leftarrow$ front $\rightarrow$ | $\leftarrow$ bleed $\rightarrow$ |

- bleed** Colour extends beyond the trim edge so no white border appears after guillotining.
- back** Back cover panel (width mm wide).
- groove** Hinge recess for hardcover binding. Filled with the panel colour; text is kept clear automatically
- spine** Width calculated from `pages` and `grammage`.
- front** Front cover panel (width mm wide).

4 Spine width calculation

$$\text{spine_width} = (\text{pages} \times \text{mm_per_sheet}) / 2$$

The division by 2 accounts for two pages per physical sheet. Example:

$$600 \text{ pages} \times 0.10 \text{ mm/sheet} \div 2 = 30 \text{ mm}$$

5 Basic structure

```
\documentclass[
  pages=600,
  grammage=75,
  colorFront=1B6B7B,
  colorSpine=D4A96A]
{bookcover}

\begin{front}
  % front cover content
\end{front}

\begin{back}
  % back cover content
\end{back}

\begin{spine}
  % spine content
\end{spine}

\begin{document}
  \makecover
\end{document}
```

`\makecover` is the only command needed inside `document`.

6 Environments

6.1 front

Defines the content of the front (right-hand) cover panel.

- Live area width: `width` mm (groove is included in the underlying minipage but hidden from text by the automatic margin).
- Automatic margins: `margin+groove` on the left (hinge side), `margin` on the right.
- Use `\colorblock` for full-width colour bands that span the groove.

```
\begin{front}
  \centering\color{white}
```

```
{\Huge\bfseries BOOK TITLE}
\vfill
{\large\bfseries AUTHOR NAME}
\vspace{4mm}
\colorblock{D4A96A}{10mm}{\centering\color{1B6B7B} Publisher}
\end{front}
```

6.2 back

Defines the content of the back (left-hand) cover panel.

- Live area width: `width` mm.
- Automatic margins: `margin` on the left, `margin+groove` on the right (hinge side).

```
\begin{back}
\vspace{10mm}
\color{white}
\setlength{\parskip}{14pt}
{\normalsize Book description goes here.}
\vfill
{\centering\includegraphics[width=0.25\linewidth]{logo.png}\par}
\hfill{\color{gold}\small\bfseries ISBN 978-0-000-00000-0}
\vspace{1cm}
\end{back}
```

6.3 spine

Defines the content of the spine strip.

- Width is calculated automatically from `pages` and `grammage`.
- No automatic margins as the spine may be too thin — layout is fully manual.
- Use `\spinerotated` to rotate text top-to-bottom.
- Use `\centering` and `\vfill` to distribute elements vertically.
- `\colorblock` is **not** available inside `spine`.

```
\begin{spine}
\centering
\vspace{4mm}
\color{teal}
\spinerotated{\normalsize\bfseries Author Name}\par
\vfill
\spinerotated{\large\bfseries Book Title}\par
\vfill
\spinerotated{\small 4th ed.}\par
\vfill
{\normalsize\bfseries Vol.\1}\par
\vspace{4mm}
\end{spine}
```

Spine width and readability: if the spine is narrower than approximately 12 mm, consider using only the title — or omitting text entirely — to avoid overcrowding.

7 Commands

7.1 `\colorblock`

Draws a full-panel-width colour band inside `front` or `back`. Unlike a plain `\colorbox`, it deliberately escapes the automatic content margins so the band always spans the complete panel width, including the groove area.

```
\colorblock{color}{height}{content}

% Parameters:
%   color    -- xcolor name or HTML hex without #
%   height   -- band height (any TeX length: mm, pt, \textheight...)
%   content  -- any LaTeX content; receives full panel \linewidth

% Examples:
\colorblock{D4A96A}{0.25\textheight}{
  \centering\color{1B6B7B}
  {\Huge\bfseries\MakeUppercase{Book Title}}
}

\colorblock{D4A96A}{10mm}{
  \centering\color{1B6B7B}{\bfseries Publisher Name}
}
```

Content inside `\colorblock` is vertically centred. If you need top or bottom alignment, use a `\minipage` with the desired `[t]` or `[b]` to wrap the content argument.

7.2 `\spinerotated`

Rotates content -90° (top-to-bottom, standard book spine direction) and centres it across the full spine width.

```
\spinerotated{text}
\spinerotated{\large\bfseries Book Title}\par
\vfill
\spinerotated{\normalsize Author Name}\par
```

Do not wrap multiple elements in a single `\spinerotated` call if you intend to use `\vfill` between them — the fill will not work inside a rotated box. Call `\spinerotated` once per element instead, and do not forget the `\par` at the end.

7.3 `\makecover`

Triggers the full TikZ render of the cover sheet. Must be the **only content** inside document.

```
\begin{document}
  \makecover
\end{document}
```

`\makecover` performs the following steps in order:

1. Converts all lengths to unitless mm values for `pgfmath`.
2. Computes horizontal and vertical zone boundaries.
3. Fills background colours (back, front, spine) with a 0.2 mm overlap between adjacent fills to prevent hairline seams in PDF viewers.
4. Draws crop marks at the four corners of the trim box.
5. Draws dimension annotations below and to the right of the artwork (for reference only — you may remove before sending to press).
6. Places content nodes for back, spine, and front panels inside clipped scopes so overflow cannot bleed across zones.

8 Automatic margin behaviour

The `margin` class option sets a horizontal content margin that is applied automatically to `front` and `back`. The groove-adjacent side always receives `margin+groove` to keep text clear of the binding hinge:

Panel	Left margin	Right margin
front	margin + groove	margin
back	margin	margin + groove

`\colorblock` bypasses these margins intentionally so that colour bands span the full panel. All other content respects them.

9 Dimension annotations

`\makecover` automatically draws labelled arrows below and to the right of the artwork showing:

- bleed, back, spine, front, bleed widths (horizontal)
- groove widths (horizontal, labelled below the arrow line)
- cover height (vertical, on the right)

These annotations are **not** part of the print-ready output. To remove them before sending to press, comment out the annotation block in `\bc@drawall` (step 3 in the source), or simply crop the PDF to the trim box using your prepress software.

10 Complete example

```
\documentclass[
  margin      = 10,
  pages       = 600,
  grammage    = 75,
  colorFront  = 1B6B7B,
  colorBack   = 1B6B7B,
  colorSpine  = D4A96A
]{bookcover}
```

```

\usepackage[english]{babel} % hyphenation patterns
\usepackage{graphicx} % \includegraphics
\usepackage{ragged2e} % \justifying
\usepackage{fontspec} % requires XeLaTeX or LuaLaTeX.
    \setmainfont{TeX Gyre Adventor}
    \newfontfamily\TitleFont{TeX Gyre Adventor}[LetterSpace=10.0]
\usepackage{xcolor}
    \definecolor{teal}{HTML}{1B6B7B} % matches colorFront / colorBack
    \definecolor{gold}{HTML}{D4A96A} % matches colorSpine
    \definecolor{white}{rgb}{1,1,1}

%% =====
%% TYPOGRAPHIC SCALE
%% =====
\def\hugefont {\fontsize{48}{56}} % main cover title
\def\largefont {\fontsize{30}{36}} % spine title
\def\mediumfont{\fontsize{20}{24}} % subtitle, edition
\def\normalfont{\fontsize{16}{19}} % authors, publisher, description
\def\smallfont {\fontsize{12}{14}} % spine edition, ISBN

%% =====
%% BOOK METADATA
%% =====
%% --- Front cover ---
\def\booktitle{Discrete\Mathematics}
\def\booksubtitle{Finally Proving P\,=\,NP (We Think)}
\def\bookedition{${\infty}$th edition}
\def\coverimage{cover_graph.pdf}
\def\bookauthors{%
    Al G. Bra
    \quad\textbullet\quad
    Kay O. S. Theory%
}
\def\bookpublisher{${\mathcal{O}}(1)$ Press}

%% --- Back cover ---
\def\bookdescription{%
    This textbook provides a rigorous introduction to discrete
    mathematics. All proofs are straightforward and left as an
    exercise to the reader.

    The second edition features twice the content of the first,
    achieved via the Banach--Tarski decomposition method.
    The publisher has been informed.

    The third part covers graph theory, including a definitive
    solution to the Bridges of K\{o}nigsberg problem.
    (We added a bridge. You're welcome.)%
}
\def\publisherlogo{logo.png}
\def\bookisbn{ISBN 978-0-314-15926-5}

```

```

%% --- Spine ---
\def\spinetitle{Discrete Mathematics}
\def\spineauthors{Bra \textbullet Theory}
\def\spineedition{${\infty}$th ed.}
\def\spinevolume{Vol.\n$}

%% =====
%% FRONT COVER
%% =====
\begin{front}

  %% Title and Subtitle - full-width band at the top
  \colorblock{gold}{0.25\textheight}{%
    \centering
    \color{teal}

    {\hugefont\TitleFont\bfseries\selectfont
      \MakeUppercase{\booktitle}\par}

    \vspace{2mm}
    {\rule{0.85\textwidth}{2pt}\par}
    \vspace{2mm}

    {\mediumfont\bfseries\selectfont
      \MakeUppercase{\booksubtitle}\par}
  }

  \centering
  \color{gold}

  %% Edition - pushed to the right margin with \hfill.
  \vspace{5mm}
  \hfill{\mediumfont\itshape\selectfont\bookedition}

  %% Cover image - fills available vertical space between edition line
  %% and author names.
  \vfill
  \includegraphics[width=\linewidth]{\coverimage}

  %% Author names - bold, uppercased, centred.
  \vfill
  {\normalfont\bfseries\selectfont\MakeUppercase{\bookauthors}}

  %% Publisher strip - another full-width band at the bottom.
  \vspace{4mm}
  \colorblock{gold}{10mm}{%
    \centering
    \color{teal}
    {\normalfont\bfseries\selectfont\bookpublisher}
  }

\end{front}

```

```

%% =====
%% BACK COVER
%% =====
\begin{back}

  \vspace{10mm}
  \color{white}
  \setlength{\parindent}{0pt}
  \setlength{\parskip}{14pt}      % breathing room between paragraphs

  %% Book description - justified text using ragged2e's \justifying,
  {\normalfont\selectfont\justifying\bookdescription}

  \vfill

  %% Publisher logo - centred, sized to 25% of the text width.
  {\centering\includegraphics[width=0.25\linewidth]{\publisherlogo}\par}

  %% ISBN - right-aligned, at the very bottom.
  \hfill{\color{gold}\smallfont\bfseries\selectfont\bookisbn}

  \vspace{1cm}
\end{back}

%% =====
%% SPINE
%% =====
\begin{spine}
  \centering
  \vspace{2mm}
  \color{teal}

  %% Authors at the top
  \spinerotated{{\normalfont\bfseries\selectfont\spineauthors}}\par

  %% Title in the centre - largest element on the spine
  \vfill
  \spinerotated{{\largefont\bfseries\selectfont\spinetitle}}\par

  %% Edition below the title
  \vfill
  \spinerotated{{\smallfont\bfseries\selectfont\spineedition}}\par

  %% Volume badge - upright text at the bottom of the spine
  \vfill
  {\centering\normalfont\bfseries\selectfont\spinevolume\par}

  \vspace{2mm}
\end{spine}

%% =====
%% DOCUMENT BODY

```

```
%% \makecover is the only content needed
%% =====
\begin{document}
\makecover
\end{document}
```